IEOR 221 - HW 6 - Jie Wen

1

The cash position X_t follows the process $dX = 0.5\,dt + 2\,dW_t$, where time t is measured in quarters.

Over 1 year ($t = 4$ quarters), $X_4 = X_0 + 0.5 \times 4 + 2 \times W_4 = X_0 + 2 + 2W_4$, where $W_4 \sim N(0, 4)$.

Thus, $X_4 \sim N(X_0 + 2, 16)$.

The probability of a negative cash position is $P(X_4 < 0) = \Phi\left(\frac{0-(X_0+2)}{4}\right) = \Phi\left(-\frac{X_0+2}{4}\right)$, where Φ is the CDF of the standard normal distribution.

To have $P(X_4 < 0) < 0.05$, solve $\Phi\left(-\frac{X_0+2}{4}\right) < 0.05$.

This requires $-\frac{X_0+2}{4} < \Phi^{-1}(0.05)$, where $\Phi^{-1}(0.05) \approx -1.64485$.

Thus, $-\frac{X_0+2}{4} < -1.64485$.

Therefore, $X_0 > 4.5794$.

The initial cash position must be higher than 4.58 million dollars (rounded to two decimal places) to ensure the probability is less than 5%. For exact equality to 5%, $X_0 = 4.58$ (approximately).

2

Difference Between the Lecture Model and Each Alternative

The lecture model, given by

$$\Delta S = \mu S \Delta t + \sigma S \Delta W,$$

describes the change in stock price S over a small time interval Δt, where μ is the constant drift rate (expected return per unit time), σ is the constant volatility, and ΔW is the increment of a Wiener process (with $\Delta W \sim N(0, \Delta t)$). This is the discrete approximation to the continuous-time Itô process

$$dS = \mu S\,dt + \sigma S\,dW_t,$$

known as **geometric Brownian motion (GBM)**. In this model, both the drift and diffusion terms are proportional to the current stock price S, meaning the process models *percentage* changes (returns) in the stock price:

$$\frac{\Delta S}{S} \approx \mu \Delta t + \sigma \Delta W.$$

The returns $\frac{\Delta S}{S}$ follow a normal distribution with mean $\mu \Delta t$ and variance $\sigma^2 \Delta t$, independent of the level of S. As shown in the lecture notes, this leads to the stock price S_t following a lognormal distribution:

$$\ln S_t \sim N\left[\ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right)t, \sigma^2 t\right],$$

with $S_t = S_0 \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t\right]$. Key properties include $E[S_t] = S_0 e^{\mu t}$ (exponential growth at rate μ) and positivity ($S_t > 0$ if $S_0 > 0$), as the multiplicative nature prevents crossing zero.

Now, consider the differences with each alternative model.

1. $\Delta S = \mu \Delta t + \sigma \Delta W_t$

This is the discrete approximation to a **generalized Wiener process** (or arithmetic Brownian motion with drift):

$$dS = \mu\,dt + \sigma\,dW_t.$$

- **Key differences**:
  - **Drift and volatility are absolute (fixed in dollar terms)**: The expected change is a constant $\mu \Delta t$, and the standard deviation of the change is a constant $\sigma\sqrt{\Delta t}$, regardless of the current price S. In contrast, the lecture model scales both by S, making them proportional to the price level (percentage-based).
  - **Distribution of S_t**: Here, $S_t = S_0 + \mu t + \sigma W_t$, so S_t follows a normal distribution $N(S_0 + \mu t, \sigma^2 t)$. This lacks the lognormal property of the lecture model, where S_t is skewed right and variance scales with $S_0^2 e^{2\mu t}(e^{\sigma^2 t} - 1)$.

- **Returns depend on S**: The percentage return $\frac{\Delta S}{S} \approx \frac{\mu \Delta t}{S} + \frac{\sigma \Delta W_t}{S}$ has drift and volatility that *decrease* as S increases, unlike the lecture model's constant μ and σ for returns.
- **Positivity not guaranteed**: Since increments are normally distributed and unbounded below, S_t can become negative (e.g., if a large negative ΔW_t overwhelms the drift). The lecture model avoids this via its multiplicative structure.
- **Economic interpretation**: This model implies the same dollar volatility for a $1 stock as for a $1000 stock, which is unrealistic—small price changes matter more for low-priced stocks.

2. $\Delta S = \mu S \Delta t + \sigma \Delta W_t$

This corresponds to the continuous process

$$dS = \mu S \, dt + \sigma \, dW_t,$$

a linear stochastic differential equation (SDE) blending geometric drift with arithmetic diffusion.

- **Key differences**:
  - **Drift is proportional to S, but volatility is absolute**: The expected percentage growth is constant at μ, like the lecture model, but the diffusion term $\sigma \Delta W_t$ is fixed in dollar terms, not scaled by S. In the lecture model, both terms scale with S, keeping percentage volatility constant.
  - **Distribution of S_t**: The solution is $S_t = S_0 e^{\mu t} + \sigma \int_0^t e^{\mu (t-s)} dW_s$, which is normally distributed around the deterministic trend $S_0 e^{\mu t}$ with variance depending on the integral (but not scaling quadratically with S_t^2). This differs from the lognormal in the lecture model, lacking the same skewness and variance structure.
  - **Returns depend on S**: $\frac{\Delta S}{S} \approx \mu \Delta t + \frac{\sigma \Delta W_t}{S}$, so percentage volatility $\frac{\sigma}{\sqrt{\Delta t} S}$ *decreases* as S rises, unlike the constant σ in the lecture model. For high S, the process behaves almost deterministically ($S_t \approx S_0 e^{\mu t}$), understating risk.
  - **Positivity not guaranteed**: The fixed-volatility term can drive S_t negative, especially if σ is large relative to S, as there's no mechanism to dampen volatility near zero. The lecture model's $ \sigma S \Delta W $ term reduces volatility as S approaches zero, preserving positivity.
  - **Economic interpretation**: This implies constant dollar volatility, so a high-priced stock has lower percentage risk than a low-priced one, which doesn't match observed stock behavior where volatility is roughly proportional to price (e.g., a 35% annual volatility applies similarly to stocks at different price levels).

3. $\Delta S = \mu \Delta t + \sigma S \Delta W_t$

This approximates the continuous process

$$dS = \mu \, dt + \sigma S \, dW_t,$$

an SDE with arithmetic drift and geometric diffusion.

- **Key differences**:
  - **Volatility is proportional to S, but drift is absolute**: The diffusion scales with S (percentage volatility constant at σ), like the lecture model, but the drift $\mu \Delta t$ is fixed in dollars, not scaled by S. The lecture model has both terms proportional to S.
  - **Distribution of S_t**: The solution is more complex: $S_t = e^{\sigma W_t - \frac{1}{2}\sigma^2 t} \left(S_0 + \mu \int_0^t e^{-\sigma W_s + \frac{1}{2}\sigma^2 s} ds \right)$. This is not lognormal; the integral introduces path-dependence, and $E[S_t] = S_0 + \mu t$ (linear growth), unlike the exponential $S_0 e^{\mu t}$ in the lecture model.
  - **Returns depend on S**: $\frac{\Delta S}{S} \approx \frac{\mu \Delta t}{S} + \sigma \Delta W_t$, so percentage drift $\frac{\mu}{S}$ *decreases* as S rises, unlike the constant μ in the lecture model. For large S, growth slows in percentage terms.
  - **Positivity**: This model is better than the first two at preserving positivity because the diffusion $\sigma S \Delta W_t$ vanishes as $S \to 0$, and with positive μ, the process tends to be pushed upward. However, it can still hit or absorb at zero under certain conditions (e.g., if $\mu \leq 0$ or volatility is high), and rigorous analysis shows it may reach zero with positive probability, unlike the strict positivity of GBM.
  - **Economic interpretation**: Constant dollar drift implies the same expected dollar growth for a $1 stock as for a $1000 stock, so low-priced stocks have unrealistically high percentage expected returns, while high-priced ones have low ones. This doesn't align with finance, where expected returns are typically modeled as percentages independent of price level.

Why the Lecture Model (Equation (*)) Is More Appropriate for Stock Price Behavior

The lecture model (GBM) is preferred because it captures key empirical and theoretical features of stock prices better than the alternatives:

- **Lognormal distribution and positivity**: Stock prices cannot be negative, and historical data shows they are approximately lognormally distributed (right-skewed, with variance increasing over time and with price level). GBM ensures $S_t > 0$ naturally through its multiplicative structure, while alternatives 1 and 2 allow negatives, and alternative 3 risks hitting zero.
- **Constant percentage returns**: Financial theory and data suggest stock *returns* ($\frac{\Delta S}{S}$) have roughly constant mean (μ) and volatility (σ), independent of the price level. This matches GBM, where returns are normally distributed. In contrast:
  - Alternative 1 makes both absolute.
  - Alternative 2 makes volatility percentage-decreasing.
  - Alternative 3 makes drift percentage-decreasing.
  These lead to unrealistic behaviors, like diminishing risk or growth for high-priced stocks.
- **Exponential growth**: GBM gives $E[S_t] = S_0 e^{\mu t}$, reflecting compounding returns (e.g., a 10% annual expected return grows a $100 stock to ~$110, and a $1000 stock to ~$1100). Alternatives deviate: linear growth in 1 and 3, or mismatched variance in 2.
- **Empirical fit and applications**: While real stocks exhibit jumps, fat tails, and volatility clustering (violating GBM assumptions), GBM is a reasonable first approximation supported by the central limit theorem for aggregated small changes. It's the foundation for Black-Scholes-Merton (as derived in the notes via Itô's lemma and risk-neutral pricing), binomial trees (as $N \to \infty$), and risk-neutral valuation, where the drift μ is replaced by r without affecting prices. The alternatives don't lead to closed-form solutions like the BSM formula and fail in these contexts (e.g., they might allow arbitrage or negative prices).
- **Economic realism**: Stocks are claims on growing firms, so percentage-based modeling aligns with relative changes mattering more than absolute ones (e.g., a $10 drop hurts a $20 stock more than a $2000 one). GBM's assumptions (no jumps, symmetric returns) are simplifications, but superior to the alternatives' mismatches in scaling.

In summary, GBM models stock *prices* as evolving multiplicatively, aligning with observed return dynamics, while the alternatives mis-scale drift or volatility, leading to implausible properties like negative prices or non-constant return statistics.

3

Let S_t follow geometric Brownian motion (GBM)

$$dS_t = \mu S_t, dt + \sigma S_t, dW_t.$$

Define $Y_t = S_t^n$. Apply Itô's lemma to $f(S) = S^n$:

$$df(S) = f'(S), dS + \tfrac{1}{2} f''(S), (dS)^2,$$

with $f'(S) = nS^{n-1}$ and $f''(S) = n(n-1)S^{n-2}$. For GBM, $(dS)^2 = \sigma^2 S^2, dt$ (the quadratic variation term). Hence

$$\begin{aligned} d(S_t^n) &= nS_t^{n-1}(\mu S_t, dt + \sigma S_t, dW_t) + \tfrac{1}{2} n(n-1) S_t^{n-2} \big(\sigma^2 S_t^2, dt\big) [2pt] \\ &= S_t^n \Big[\big(n\mu + \tfrac{1}{2} n(n-1)\sigma^2\big), dt + n\sigma, dW_t \Big]. \end{aligned}$$

This has the GBM form

$$dY_t = \tilde{\mu}, Y_t, dt + \tilde{\sigma}, Y_t, dW_t \quad \text{with} \quad \tilde{\mu} = n\mu + \tfrac{1}{2} n(n-1)\sigma^2, ; ; \tilde{\sigma} = n\sigma,$$

so S_t^n is itself a geometric Brownian motion (with drift $\tilde{\mu}$ and volatility $\tilde{\sigma}$). The Itô rule and the GBM definition used above are standard; see references on Itô's lemma and GBM.

4

For GBM $dS = \mu S, dt + \sigma S, dW_t$:

$$S_{t+\Delta t} = S_t \exp\left((\mu - \tfrac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}, Z \right)$$

with $Z \sim N(0,1)$.
Hence

$$\mathbb{E}[S_{t+\Delta t}] = S_t e^{\mu \Delta t}, \quad \mathrm{Var}(S_{t+\Delta t}) = S_t^2 e^{2\mu\Delta t} \left(e^{\sigma^2 \Delta t} - 1 \right),$$

and any percentile comes from the lognormal quantile above.

Inputs: $S_0 = 50,\ \mu = 0.16,\ \sigma = 0.30,\ \Delta t = \frac{1}{250} = 0.004.$

(a) Expected price tomorrow

$$\mathbb{E}[S_1] = 50, e^{0.16/250} = 50.0320.$$

(b) Standard deviation of tomorrow's price

$$\mathrm{SD}(S_1) = \sqrt{50^2 e^{2(0.16)/250} \left(e^{0.30^2/250} - 1 \right)} = 0.9494.$$

(c) 95% confidence limits for tomorrow's price

Use the 2.5% and 97.5% lognormal quantiles:

$$S_{\text{low}} = 50 \exp\left((\mu - \tfrac{1}{2}\sigma^2)\Delta t - 1.96, \sigma\sqrt{\Delta t} \right),$$

$$S_{\text{high}} = 50 \exp\left((\mu - \tfrac{1}{2}\sigma^2)\Delta t + 1.96, \sigma\sqrt{\Delta t} \right),$$

with $(\mu - \frac{1}{2}\sigma^2)\Delta t = (0.16 - 0.045)\cdot 0.004 = 0.00046,\ \sigma\sqrt{\Delta t} = 0.30\sqrt{0.004} = 0.01897.$
This gives

$$[48.1969,\ 51.9183].$$